

# Natural Computing



LUDWIG-  
MAXIMILIANS-  
UNIVERSITÄT  
MÜNCHEN

LMU Munich  
summer term 2025

Thomas Gabor



# Werkstudent Softwareentwickler (m/w/d)

## Quantum Computing, Web & KI

### Deine Aufgaben:

- Software- & Schnittstellenentwicklung (REST-APIs, QC-Hardware)
- Frontend-Visualisierung mit React.js
- DevOps & Deployment (Docker, Kubernetes)
- Analyse & Entwurf einer Domänensprache (DSL)
- Hybride Algorithmen (Python, klassisch + quantenbasiert)

### Dein Profil:

- Laufendes Studium IT/Physik/Mathe/Ing.
- Interesse Quantencomputing & Kommunikationstech.
- Erste Erfahrung Python/JavaScript, React.js+, REST, Docker/K8s von Vorteil

### Wir bieten:

- ✓ 100% Remote
- ✓ Teilzeit
- ✓ Flexible Arbeitszeiten

**Ansprechpartner:**  
Tobias Rohe



AIGNOSCO GmbH, München



# Exam Date

Tuesday, July 29, 2025

start 9:00, duration 90 minutes

 *probably* 9:30

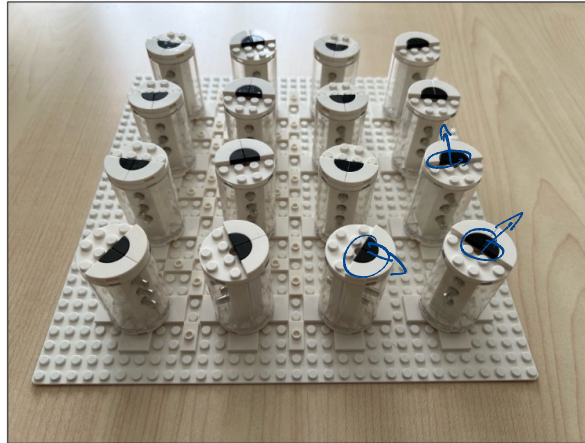
Großhaderner Str. 9, room N 00.001

details and registration soon in moodle

Previously on...



## Experiment Time!



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# Ising-QUBO Equivalence

$J, h$

$$\sigma \in \{-1, +1\}^n$$

$Q$

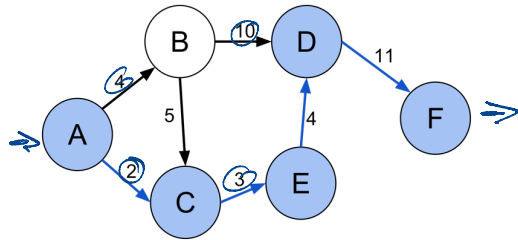
$$x \in \{0, 1\}^n$$

$$\sigma_i = 2x_i - 1$$

# Metaheuristic Search and Optimization



# Shortest Path



example

$$x = \{ \underline{(A, C)}, \underline{(C, E)}, \underline{(E, D)}, \underline{(D, F)} \}$$

$$\mathcal{X} = \rho(E) = 2^E$$

for graph edges  $E$

$$\mathcal{T} = \mathbb{Z}$$

$$\tau = \{e_1, \dots, e_n\} \mapsto \sum_{i=1}^n \text{cost}(e_i)$$

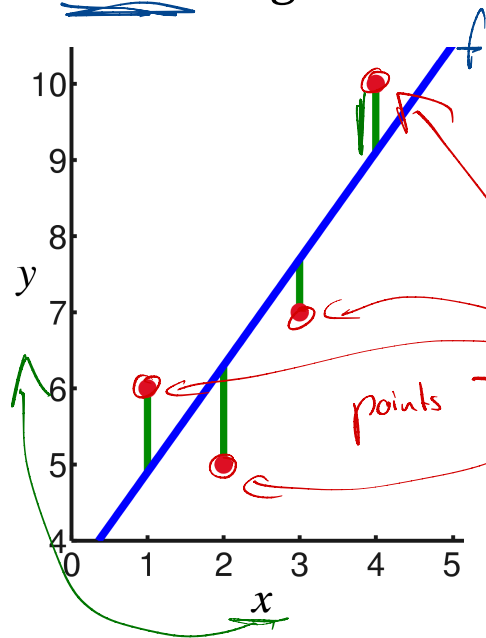
$$\tau: \mathcal{X} \rightarrow \mathcal{T}$$

$$x^* = \underset{x \in \mathcal{X}}{\text{argmin}} \tau(x)$$

source:

[en.wikipedia.org/wiki/Shortest\\_path\\_problem#/media/File:Shortest\\_path\\_with\\_direct\\_weights.svg](https://en.wikipedia.org/wiki/Shortest_path_problem#/media/File:Shortest_path_with_direct_weights.svg)

# Linear Regression



points  $T = \{(x_1, y_1), \dots, (x_n, y_n)\}$

$$\mathcal{X} = \mathbb{R}^2$$

so that

$$f(x) = \underline{ax} + \underline{b} \quad \text{for } \underline{x} = (a, b)$$

$$\mathcal{Y} = \mathbb{R}$$

$$\tau(a, b) = \sum_{i=0}^{n-1} |(ax_i + b) - y_i|$$

$$\mathcal{F} = \{f: \mathbb{R} \rightarrow \mathbb{R} \mid f \text{ is linear}\}$$

↳ mathematically possible  
but very inefficient

source:

[en.wikipedia.org/wiki/Linear\\_regression#/media/  
File:Linear\\_least\\_squares\\_example2.png](https://en.wikipedia.org/wiki/Linear_regression#/media/File:Linear_least_squares_example2.png)

# Eierkartons!



source: Wikipedia

Wir brauchen Eierkartons...